Cardano Infrastructure Topology

Concentration findings for linear-Leios EB voting

Snapshot date: 2026/05/11 · Source: Koios *pool_list* + curated MPO mapping · Scope: **2,915** active pools / **21.15B ADA** covering the full Cardano network (**84** MPO entities + **2,258** single-pool operators)

Executive summary

This pass resolves every registered relay endpoint across the full Cardano pool population to its hosting provider, ASN, country, and cloud region. The motivation is the linear-Leios EB voting rule from CIP-0164: with a high certification threshold (illustrated at τ=75% in the CIP), the spec itself acknowledges that "a 26% stake attacker can trivially attack Leios throughput." The relevant concentration question is therefore whether any single failure domain encompasses ≥26% of total productive stake.

No single cloud region holds more than ~5% of total productive stake. The largest regional concentration is **GCP europe-west3 (Frankfurt) at 4.67%**, dominated by Coinbase. AWS ap-northeast-1 (Tokyo) follows at 4.06%; ap-northeast-2 (Seoul), eu-north-1 (Stockholm), eu-west-1 (Ireland), and Azure westeurope (Amsterdam) all sit between 2.5 - 3.2%. A single-region outage cannot by itself threaten the 75% EB quorum.

Provider-level shares are higher: **AWS 22.7%, OVH 11.9%, GCP 11.6%, Hetzner 6.0%, Contabo 4.7%, Azure 4.0%, DigitalOcean 2.4%** of total productive stake. **Complete AWS unavailability silences 22.84% of stake — sitting only 3.16 points below the 26% spec-grounded threshold.** A correlated AWS+GCP outage silences 35.48% and crosses the threshold by 9.48 points. The full joint-hyperscaler scenario (AWS+GCP+Azure) silences 41.75%. The failure-scenarios section below quantifies eleven concrete plausible events.

Coverage: **2,915 active pools / 21.15B ADA** — the entire active Cardano pool population, including both the 85 MPO entities (~16.4B ADA) and the long tail of single-pool operators (~5.3B ADA). All percentages in this report are computed against this full base. The 6.7% "Unresolved" share is pools whose registered relays no longer resolve (stale DNS or decommissioned endpoints).

Concentration at a glance

Two single-number summaries are useful before diving in. The **Herfindahl-Hirschman Index** (HHI) is the sum of squared stake-share fractions, scaled to 0–10,000 — the antitrust convention reads HHI < 1,500 as low concentration, 1,500–2,500 as moderate, > 2,500 as high. The **effective number of domains** is exp(Shannon entropy) and reads as "how many equally-sized domains would give the same diversity as what is actually observed".

Dimension	Distinct labels	HHI	Effective ■ of domains	Top-1 share	Concentration grade
provider	30	1267.4	10.97	22.71%	low concentration
region	160	448.8	38.16	12.73%	low concentration
physical_country	66	1074.1	13.89	20.43%	low concentration
tier	6	2999.5	3.79	40.11%	high concentration

The provider, region, and physical-country dimensions all read as low-concentration in the antitrust sense — ~11 effective providers and ~38 effective regions. The **tier** dimension is the only high-concentration aggregate: just ~3.8 effective tiers, because over 97% of stake is split between only three structural layers (hyperscaler, commodity-VPS, on-prem). This is the shared-fate posture in a single number.

Hyperscaler vs commodity vs residential

Cardano's full active pool population splits roughly 40% / 31% / 19% across hyperscaler / commodity-VPS / on-prem-or-unclassified, with a small residential tail. Adding the long tail of single-pool operators on top of the MPO base lowers the hyperscaler share (single-pool operators run far more on Hetzner, Contabo, OVH, or self-hosted hardware than on AWS / GCP / Azure) and grows the commodity-VPS slice. The 19% "other / on-prem" is dominated by small ASNs that don't map to a recognised commercial hoster — bare-metal in residential or business broadband, regional hosters, and operators behind generic colocation.

Tier	Stake (B ADA)	Share	Entities
hyperscaler	8.48B	40.1%	469
commodity-vps	6.61B	31.2%	868
other-or-onprem	4.02B	19.0%	816
unresolved	1.42B	6.7%	419
residential/satellite	0.61B	2.9%	14
edge/cdn	0.01B	0.0%	6

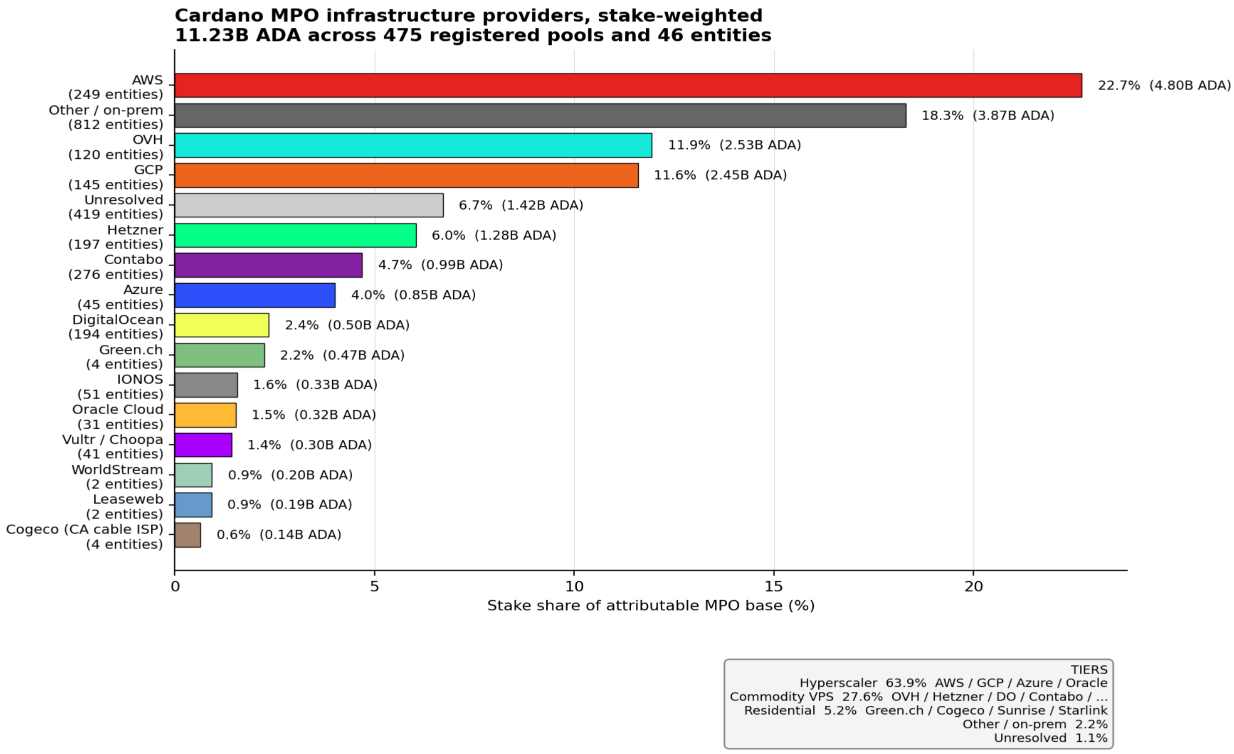

Figure 1. Stake-weighted infrastructure provider concentration across the full Cardano pool population. AWS, OVH, GCP, Hetzner, and Contabo are the top five.

Single-provider exposure

A complete AWS outage would silence ~23% of total productive stake — sitting right at the 25% threshold for EB quorum. A coordinated AWS + GCP outage (~34%) would cross it comfortably. OVH at 12% is the largest single-provider exposure outside the hyperscalers — a France-wide OVH incident would matter, though OVH's stake is more geographically dispersed across Roubaix, Strasbourg, Beauharnois, and Frankfurt.

Provider	Stake (B ADA)	Share	Entities
AWS	4.80B	22.7%	249
Other / on-prem	3.87B	18.3%	812
OVH	2.53B	11.9%	120
GCP	2.45B	11.6%	145
Unresolved	1.42B	6.7%	419
Hetzner	1.28B	6.0%	197
Contabo	0.99B	4.7%	276
Azure	0.85B	4.0%	45
DigitalOcean	0.50B	2.4%	194
Green.ch	0.47B	2.2%	4
IONOS	0.33B	1.6%	51
Oracle Cloud	0.32B	1.5%	31
Vultr / Choopa	0.30B	1.4%	41
WorldStream	0.20B	0.9%	2
Leaseweb	0.19B	0.9%	2
Cogeco (CA cable ISP)	0.14B	0.6%	4

Hyperscaler region exposure — the Leios-specific cut

PTR records expose the AWS region directly; GCP and Azure regions are derived from the providers' published IP-range manifests (Google's *cloud.json* and Microsoft's *ServiceTags_Public* JSON). This gives per-region exposure for every hyperscaler endpoint. **No single cloud region holds more than 5% of total productive stake.** The largest is GCP europe-west3 (Frankfurt), dominated by Coinbase, at 4.67%.

Three observations: (1) The top regional concentrations cluster around Frankfurt, Tokyo, Seoul, Stockholm, Ireland, and Amsterdam — each within a couple percentage points of the others, each dominated by one or two entities. A region-level outage is primarily a *single-entity* event, not a multi-tenant collapse. (2) Coinbase deliberately spans **six distinct hyperscaler regions across two providers** (AWS Tokyo / Ireland / Frankfurt; GCP Frankfurt and others) — its Leios-vote liveness is structurally the most resilient. (3) The fragile single-region operators are still the same names from the MPO-only cut: **Upbit** sits almost entirely in AWS Seoul (ap-northeast-2), **CHUCK BUX** mostly in AWS Stockholm (eu-north-1), and **eToro** entirely in Azure westeurope.

Hyperscaler region	Stake (B ADA)	Share	Entities
gcp:europe-west3	0.99B	4.7%	6
aws:ap-northeast-1	0.86B	4.1%	24

Hyperscaler region	Stake (B ADA)	Share	Entities
aws:ap-northeast-2	0.67B	3.2%	4
aws:eu-north-1	0.63B	3.0%	2
aws:eu-west-1	0.62B	2.9%	15
azure:westeurope	0.52B	2.5%	10
aws:eu-central-1	0.41B	2.0%	20
gcp:europe-west2	0.31B	1.4%	4
gcp:europe-west4	0.25B	1.2%	8
aws:ap-south-1	0.23B	1.1%	9
gcp:asia-northeast1	0.22B	1.1%	8
azure:eastus2	0.21B	1.0%	8
gcp:us-east4	0.20B	1.0%	2
aws:ap-southeast-2	0.18B	0.9%	10
aws:us-east-2	0.16B	0.8%	67
gcp:asia-east2	0.14B	0.7%	3
gcp:us-east1	0.13B	0.6%	11
gcp:us-central1	0.05B	0.2%	75

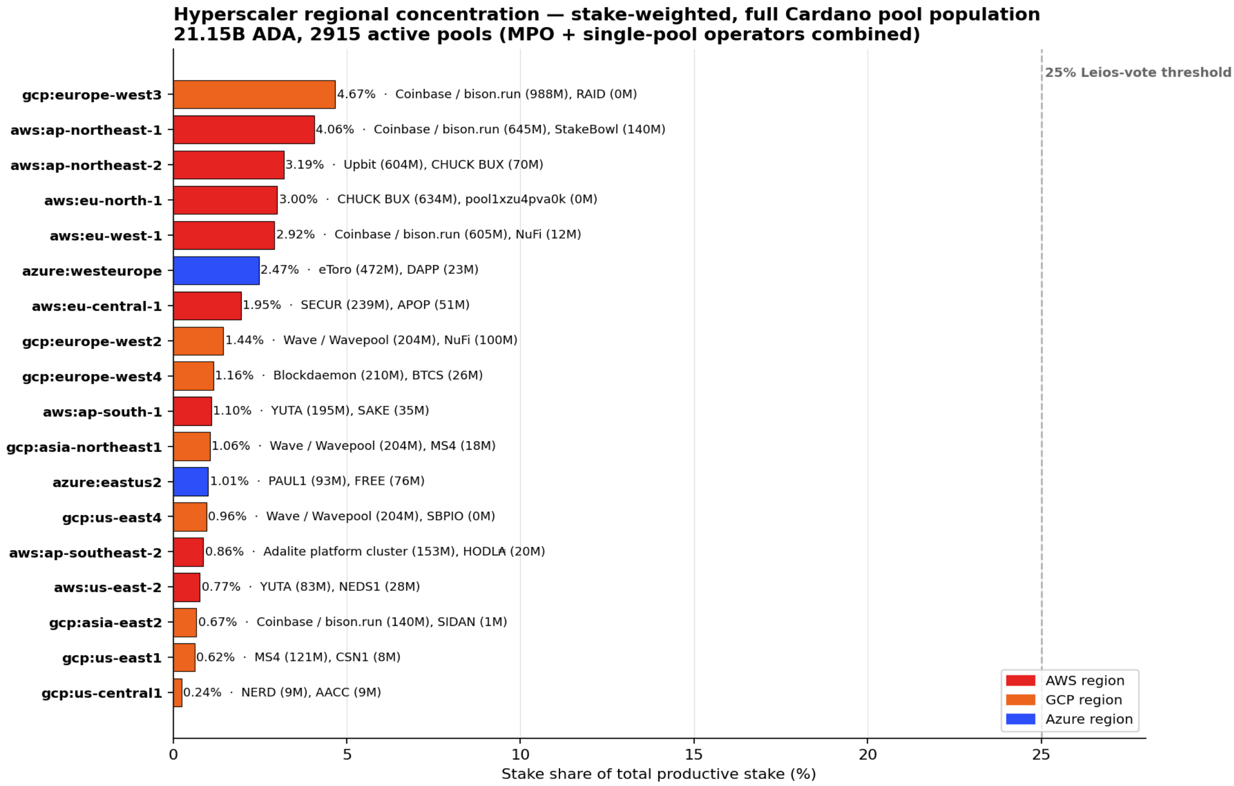

Figure 2. Hyperscaler region stake exposure with dominant entity per region. AWS in red, GCP in orange, Azure in blue. The dashed grey line marks the 26% spec-grounded threshold (CIP-0164) — all observed regions sit well below.

Geographic country distribution (physical datacentre)

The country aggregate below uses physical-datacentre country, derived from the cloud region (e.g., *aws:eu-west-1* → IE, *gcp:europe-west3* → DE, *azure:westeurope* → NL) for hyperscaler endpoints and from MaxMind GeoLite2-City for commodity-VPS endpoints. **Germany is the largest single physical country** at 19.9% because Frankfurt is a major hyperscaler / OVH / Hetzner hub. The United States follows at 19.2%; France at 11.0% reflects OVH's home country.

Physical country	Stake (B ADA)	Share	Entities
DE	4.32B	20.4%	554
US	4.06B	19.2%	892
FR	1.86B	8.8%	113
JP	1.25B	5.9%	90
GB	1.15B	5.5%	85
NL	1.14B	5.4%	58
SE	1.01B	4.8%	18
KR	0.72B	3.4%	20
IE	0.64B	3.0%	20
CA	0.60B	2.8%	97
CH	0.58B	2.8%	27
AT	0.55B	2.6%	81

Cross-checks against curated MPO profiles

The diagnostic repo carries hand-curated profiles for the largest MPO entities. Below, each profile's self-disclosed posture is set against what the topology pass actually observes on the public-relay surface.

Entity	Curated profile says	Topology pass observes	Verdict
Coinbase	Hidden behind bison.run / herd.run	51% AWS (Tokyo, Ireland, Frankfurt) + 46% GCP (europe-west3)	Six hyperscaler regions, two providers: most resilient observed
Everstake	Multi-cloud, geo-distributed bare-metal	OVH 33% / WorldStream 33% / Leaseweb 33%	Multi-provider bare-metal: confirmed
Cardano Foundation	(not in curated profiles)	100% Green.ch (Swiss bare-metal)	Single-provider, Swiss-resident
Kiln	Institutional, dedicated infrastructure	95% OVH	Single-provider, French
Blockdaemon	Multi-cloud, SOC 2 / ISO 27001	100% GCP on the registered relays	Profile claims wider scope than relays expose
Emurgo	(IOG sister entity)	97% OVH	Single-provider, French
Figment	Institutional staking, 70+ PoS networks	38 SRV records under **.cardano.aeq5f.com*; none resolve in public DNS	Genuine DNS dead-end (intentional opaque routing)

Two cross-check gaps are worth flagging. Blockdaemon's curated profile states multi-cloud infrastructure, but its registered Cardano relays all PTR-resolve under **.bc.googleusercontent.com* — either the broader infrastructure runs producers that are masked behind the GCP-facing relays, or the Cardano product line specifically runs on GCP. Figment, after a fresh Koios re-harvest, is confirmed to register 38 SRV records pointing at **.cardano.aeq5f.com* hosts that resolve to nothing in public DNS — this is a deliberate opaque-routing posture, not a data-collection gap, and Figment's 923M ADA is therefore unattributable from public-relay data alone.

Sensitivity to the stake-attribution rule

Every percentage in this report uses an **even-split** rule: a pool that registers relays at AWS and Hetzner contributes 50/50 to each, so category totals sum to ~100%. To verify the headline numbers are not artefacts of that rule, the same metrics are re-computed under two alternatives: **majority** (the entire pool's stake is assigned to the most frequent label among its relays) and **any-exposure** (the pool's stake is counted in *every* distinct label it touches, so totals exceed 100% — useful for "how much stake is at risk if X goes down").

Dimension	Key	Even-split	Majority	Any-exposure
tier	hyperscaler	40.1%	39.8%	41.7%
tier	commodity-vps	31.2%	34.0%	36.1%
tier	other-or-onprem	19.0%	16.9%	23.7%
tier	unresolved	6.7%	6.5%	7.0%
tier	residential/satellite	2.9%	2.9%	2.9%
provider	AWS	22.7%	22.7%	22.8%
provider	Other / on-prem	18.3%	16.4%	23.3%
provider	OVH	11.9%	14.6%	16.4%
provider	GCP	11.6%	11.1%	12.7%
provider	Unresolved	6.7%	6.5%	7.0%
provider	Hetzner	6.0%	6.6%	8.1%
provider	Contabo	4.7%	6.5%	7.9%
provider	Azure	4.0%	4.5%	4.5%
provider	DigitalOcean	2.4%	2.3%	2.6%
provider	Green.ch	2.2%	2.2%	2.2%
provider	IONOS	1.6%	1.0%	2.4%
provider	Oracle Cloud	1.5%	1.4%	1.8%
provider	Vultr / Choopa	1.4%	1.4%	1.5%

Headline numbers are robust: tier shares move by less than three points across rules; AWS provider share is stable around 22-23%; OVH grows from 12% (even-split) to 16% (any-exposure) because OVH-multi-relay pools also touch other providers. The any-exposure column is the appropriate read for the Leios **liveness** question: even though only 22.7% of stake "belongs" to AWS under even-split, 22.8% is **exposed** to an AWS outage under any-exposure — essentially identical, meaning few multi-relay pools mix AWS with non-AWS infrastructure as a real diversification.

Per-entity provider profile (top 18 by stake)

For each MPO entity, the table shows its dominant provider and region, its provider diversity (Shannon entropy in nats — 0 = single provider, log(N) = perfectly even across N), and the top provider mix. High diversity values (Coinbase, NuFi, Everstake) indicate operators with deliberate multi-cloud or multi-provider postures; zero values (Upbit, eToro, Cardano Foundation, YUTA, Wave, Adalite) are single-provider operators.

Entity	Pools	Stake (M ADA)	Dominant provider	Dominant region	Diversity H_{prov}	Provider mix (top)
Coinbase / bison.run	48	2448	AWS (51.1%)	gcp:europe-west3 (40.3%)	0.802	AWS=51%; GCP=46%; Unresolved=3%; Other / on-prem=0%
Figment	37	923	Unresolved (100.0%)	cc:?? (100.0%)	0.000	Unresolved=100%
CHUCK BUX	17	780	AWS (90.2%)	aws:eu-north-1 (81.3%)	0.320	AWS=90%; Unresolved=10%
Binance	53	655	AWS (90.8%)	cc:US (100.0%)	0.307	AWS=91%; Oracle Cloud=9%
Kiln	11	644	OVH (94.1%)	cc:FR (62.7%)	0.225	OVH=94%; Unresolved=6%
Wave / Wavepool	17	611	GCP (100.0%)	gcp:asia-northeast1 (33.3%)	0.000	GCP=100%; Unresolved=0%
Upbit	20	604	AWS (100.0%)	aws:ap-northeast-2 (100.0%)	0.000	AWS=100%
Everstake	15	582	WorldStream (33.3%)	worldstream:naaldwijk (25.0%)	1.099	WorldStream=33%; OVH=33%; Leaseweb=33%
eToro	21	472	Azure (100.0%)	azure:westeurope (100.0%)	0.000	Azure=100%; GCP=0%
YUTA	25	452	AWS (100.0%)	aws:ap-south-1 (43.2%)	0.000	AWS=100%
Cardano Foundation	6	396	Green.ch (100.0%)	cc:CH (100.0%)	0.000	Green.ch=100%
NORTH	5	329	Other / on-prem (100.0%)	cc:SE (100.0%)	0.000	Other / on-prem=100%
NuFi	18	312	GCP (32.1%)	contabo:lauterbourg (32.1%)	1.219	GCP=32%; Cherry Servers=32%; Contabo=32%; AWS=4%
Blockdaemon	7	293	GCP (84.7%)	gcp:europe-west4 (71.7%)	0.427	GCP=85%; Unresolved=15%
1PCT	30	270	OVH (99.9%)	cc:FR (99.9%)	0.008	OVH=100%; Contabo=0%
ADV	4	261	Other / on-prem (100.0%)	cc:GB (100.0%)	0.000	Other / on-prem=100%
Emurgo	14	259	OVH (96.8%)	cc:FR (61.1%)	0.146	OVH=97%; Hetzner=3%; Huawei Cloud=0%; Alibaba Cloud=0%
SECUR	5	239	AWS (100.0%)	aws:eu-central-1 (100.0%)	0.000	AWS=100%

Joint-failure scenarios — spec-grounded threshold analysis

CIP-0164 defines an EB as certified when the stake-weighted sum of committee votes exceeds a high threshold τ (illustrated at 75% in the CIP). The CIP's own security analysis states verbatim: *"a 26% stake attacker can trivially attack Leios throughput with a 75% certification threshold."* The operational reading: if any failure domain encompasses ≥26% of total productive voting stake, EB certification can be stalled. The table below quantifies eleven concrete plausible failure scenarios, each evaluated under an "any-exposure" rule — a pool is counted as at risk if *any* of its registered relays sits in the scenario's footprint. Scenarios marked BREACH cross the 26% line.

Scenario	Stake at risk	% of total	Gap to 26%	Pools	Top entities at risk
Joint hyperscaler outage (AWS + GCP + Azure)	8.83B ADA	41.75%	**BREACH -15.75**	731	Coinbase / bison.run (2378M); CHUCK BUX (704M); Binance (655M); Wave / Wavepool (611M); Upbit (604M)
Correlated AWS + GCP outage	7.50B ADA	35.48%	**BREACH -9.48**	622	Coinbase / bison.run (2378M); CHUCK BUX (704M); Wave / Wavepool (611M); Upbit (604M); Binance (595M)
Complete AWS unavailability (control plane + all regions)	4.83B ADA	22.84%	+3.16	403	Coinbase / bison.run (1250M); CHUCK BUX (704M); Upbit (604M); Binance (595M); YUTA (452M)
Complete OVH outage (multi-datacentre)	3.48B ADA	16.45%	+9.55	228	Kiln (644M); Everstake (582M); 1PCT (270M); Emurgo (258M); nan (229M)
GCP global IAM / control-plane outage	2.69B ADA	12.71%	+13.29	221	Coinbase / bison.run (1128M); Wave / Wavepool (611M); NuFi (300M); Blockdaemon (249M); MS4 (139M)
OVH Strasbourg datacentre fire (SBG-style)	2.62B ADA	12.37%	+13.63	161	Kiln (644M); Everstake (582M); 1PCT (270M); nan (229M); Emurgo (170M)
Frankfurt metro-area outage (DE-CIX peering point)	2.39B ADA	11.31%	+14.69	83	Coinbase / bison.run (988M); Everstake (582M); SECUR (239M); AdaOcean (185M); NEDS1 (112M)
Complete Hetzner outage	1.72B ADA	8.14%	+17.86	223	EDEN (162M); TITAN (137M); Spire (103M); HOPE (78M); LCP (68M)
Single-region worst case (GCP europe-west3 / Frankfurt)	0.99B ADA	4.67%	+21.33	28	Coinbase / bison.run (988M); RAID (0M); VKP (0M); pool1nq0jx33p0 (0M); pool190vx20f4p (0M)
Azure westeurope (Amsterdam) regional outage	0.65B ADA	3.07%	+22.93	29	eToro (472M); DAPP (135M); BANDA (14M); BITA (14M); pool1cc76kmtcp (8M)
AWS us-east-1 control-plane cascade	0.26B ADA	1.23%	+24.77	77	NEDS1 (112M); YUTA (83M); COSMC (26M); VAMP (16M); pool1lq7t0qg27 (7M)

Two scenarios breach the spec-grounded threshold: a joint hyperscaler outage (AWS+GCP+Azure, 41.75% at risk) and a correlated AWS+GCP event (35.48%). Complete AWS unavailability alone sits at 22.84% — **only 3.16 points below the 26% line**, so any concurrent failure in a second cloud or in OVH pushes Cardano into the contested zone. Single-region outages (GCP europe-west3 at 4.67%, Azure westeurope at 3.07%) and historical-precedent events (OVH SBG fire at 12.37%) all sit comfortably below.

Threshold sensitivity — how the picture shifts if τ moves

CIP-0164 leaves τ as a governance-tunable parameter. The 75% illustrative value yields a 26% adversarial threshold (the spec's own "26% stake attacker" figure). A lower τ relaxes the adversarial threshold; a higher τ tightens it. The table below re-evaluates every scenario across τ ∈ {60%, 70%, 75%, 80%, 90%}. **BREACH** means the scenario silences at least (100–τ)% of stake; green margins show the remaining safety distance.

Scenario	τ=60% (adv 40%)	τ=70% (adv 30%)	τ=75% (adv 25%)	τ=80% (adv 20%)	τ=90% (adv 10%)
All hyperscalers joint	BREACH	BREACH	BREACH	BREACH	BREACH
AWS + GCP correlated	+4.52	BREACH	BREACH	BREACH	BREACH
Complete AWS outage	+17.16	+7.16	+2.16	BREACH	BREACH
Complete OVH outage	+23.55	+13.55	+8.55	+3.55	BREACH
GCP global IAM outage	+27.29	+17.29	+12.29	+7.29	BREACH
OVH SBG-style fire	+27.63	+17.63	+12.63	+7.63	BREACH
Frankfurt metro-area outage	+28.69	+18.69	+13.69	+8.69	BREACH
Complete Hetzner outage	+31.86	+21.86	+16.86	+11.86	+1.86
GCP europe-west3 (worst region)	+35.33	+25.33	+20.33	+15.33	+5.33
Azure westeurope outage	+36.93	+26.93	+21.93	+16.93	+6.93
AWS us-east-1 cascade	+38.77	+28.77	+23.77	+18.77	+8.77

Three takeaways from the sensitivity. At a conservative τ=60%, only a perfectly correlated joint-hyperscaler event breaches; the network has substantial liveness margin. At the CIP's illustrative τ=75%, complete AWS alone is 2.2 points from breach. At τ=80%, **complete AWS unavailability alone breaches** — Cardano becomes liveness-fragile to a single-cloud failure. At τ=90%, every provider-level outage breaches and even Hetzner alone (1.9 points over) crosses the line. The choice of τ therefore directly trades off Leios safety against single-provider liveness risk.

Implications for linear-Leios EB voting

Three structural observations follow from the scenarios table.

Same-region outage (one hyperscaler region)

Worst case is GCP europe-west3 (Frankfurt) at 4.67% — well below 26%. Leios EB voting continues. The affected entity itself halts — Upbit if Seoul (aws:ap-northeast-2) falls, eToro if Azure westeurope falls — but the network does not. Region-level concentration is therefore a single-entity reputational risk, not a network-liveness risk.

Single-provider outage (all of AWS)

AWS alone strips 22.84% of total productive stake — the closest single failure to the 26% line. The 3-point margin is thin enough that any concurrent failure (an OVH SBG-style fire while AWS is down, or a partial GCP regional outage) tips the network into a stall regime. This is the operationally most fragile single-cloud exposure in the data.

Correlated hyperscaler outage (the actual threat model)

An AWS+GCP correlated event (e.g., shared DNS / IAM / certificate-authority upstreams) silences 35.48% of stake and breaches the 26% threshold by 9.48 points. A full hyperscaler joint event (AWS+GCP+Azure) silences 41.75% and breaches by 15.75 points. AWS us-east-1 cascades have historically degraded GCP and Azure operations through shared upstreams, so this is not a hypothetical — it is the dominant structural risk visible in the data.

The relevant risk metric is therefore not regional concentration but the **shared-fate footprint**: roughly two-fifths of total productive stake runs on top of three public-cloud control planes whose historical failure modes are correlated through shared DNS, certificate, BGP, and IAM upstreams.

MPO vs single-pool: the asymmetric cohort risk

The headline percentages mix institutional MPO stake (15.76B ADA across 85 entities) with the long tail of single-pool operators (5.39B ADA across 2,072 entities). The two cohorts cluster on radically different infrastructure. The MPO cohort is hyperscaler-heavy — AWS, GCP, OVH, Azure dominate. The single-pool cohort is commodity-VPS- and self-hosted-heavy — the “Other / on-prem” category alone holds 43.6% of single-pool stake, mostly small regional hosters and home-broadband bare-metal.

Provider	MPO cohort	Single-pool cohort	Asymmetry
AWS	28.3% (4464M ADA)	6.3% (339M ADA)	+22.0 pts — MPO-heavy
Other / on-prem	9.6% (1520M ADA)	43.6% (2351M ADA)	-34.0 pts — SPO-heavy
OVH	13.2% (2075M ADA)	8.4% (452M ADA)	+4.8 pts — balanced
GCP	14.7% (2313M ADA)	2.6% (139M ADA)	+12.1 pts — MPO-heavy
Unresolved	7.5% (1188M ADA)	4.3% (230M ADA)	+3.3 pts — balanced
Hetzner	4.1% (644M ADA)	11.7% (631M ADA)	-7.6 pts — SPO-heavy
Contabo	3.2% (498M ADA)	9.2% (494M ADA)	-6.0 pts — SPO-heavy
Azure	4.7% (748M ADA)	1.9% (101M ADA)	+2.9 pts — balanced
DigitalOcean	2.3% (366M ADA)	2.4% (131M ADA)	-0.1 pts — balanced
Green.ch	2.5% (397M ADA)	1.4% (77M ADA)	+1.1 pts — balanced
IONOS	1.3% (206M ADA)	2.3% (122M ADA)	-1.0 pts — balanced
Oracle Cloud	1.2% (183M ADA)	2.6% (141M ADA)	-1.4 pts — balanced

The asymmetry has a sharp implication. **Within the MPO cohort alone, AWS holds 28.3%** — already over the 26% threshold derived from CIP-0164's illustrative τ=75%. The fact that single-pool operators run on a different infrastructure stack drags the network-wide AWS share down to 22.7%, but only because long-tail SPOs sit on Hetzner, Contabo, and home hardware. Any future shift of single-pool operators into hyperscaler hosting would directly erode this buffer.

Read the other way round: the long tail of single-pool operators is a structurally valuable *diversification* for Cardano's liveness posture, not a decentralisation cost. The smallest operators are precisely the ones not on AWS.

Design options for Leios + V2 incentives

Five options grounded in the data above, ranging from least to most invasive.

1. Choose τ with an explicit single-cloud safety target

The τ-sensitivity table reads as a direct trade-off. Setting τ=70% gives Cardano a 7.2-point margin against complete-AWS outage; τ=80% reduces it to a BREACH. The CIP's illustrative τ=75% leaves a 2.2-point margin — tight. Governance should pick τ explicitly in awareness of the AWS dependency, not as a free parameter.

2. Shared-fate-aware committee sortition

CIP-0164's persistent + non-persistent voter sortition is currently stake-weighted and infrastructure-blind. A version that down-weights candidate voters whose registered relays cluster on already-represented (provider, region) pairs would produce committees whose failure modes are *de-correlated* by construction.

This is a sortition modification only — no protocol-level change to the incentive layer.

3. V2 reward modulation against hyperscaler concentration

If the V2 reward formula incorporates an infrastructure-diversity coefficient (stake-weighted bonus for pools whose registered relays span ≥2 providers, or penalty for top-1-provider concentration above a threshold), the incentive to diversify becomes direct. The diversity Shannon-entropy H computed in the profile-diff table is the natural metric.

4. Transparency requirement on producer-node location

The 22.84% AWS exposure is computed from registered relays only. The actual producer location is hidden by every serious operator, so the true concentration could be higher or lower. A protocol-level requirement to declare the producer's ASN (in the pool registration certificate, hashed if privacy-sensitive) would let future analyses ground in producer reality rather than relay proxy. This is a minor on-chain schema addition.

5. Reverse-decentralisation incentive: protect the long tail

The single-pool cohort's 43.6% “Other / on-prem” share is the most valuable diversification Cardano has. The V2 design should explicitly recognise this and avoid changes that would push small operators into hyperscaler hosting (e.g., per-block compute requirements they can only afford on managed cloud). The current pledge-bonus-forgoing equilibrium in the MPO cohort is unrelated to this structural diversification — the two should be addressed independently.

Appendix — Sources and methodology

Executive summary

The analysis combines four classes of data: live public-API fetches performed at analysis time, the diagnostic repository's existing curated MPO census, well-established public IP-attribution conventions encoded into the classifier, and a small number of standard cloud-region geography facts. Every concrete number in the tables and figures traces back to either a live fetch (with the raw response cached on disk) or a file in the diagnostic repository. The pipeline is fully reproducible: re-running *fetch_koios_pool_list.py* followed by *build_mpo_topology_concentration.py* against the same Koios snapshot reconstructs the analysis end to end.

Live data fetches performed at analysis time

The active-pool population and its registered relay set come from the public Koios */pool_list* endpoint — 6,133 pool records covering the full Cardano network, of which 2,915 are active. No authentication, no rate limit hit, response cached to *data/koios_pool_list.json*. This is the source of the headline 2,915 pools / 21.13B ADA figures.

Forward DNS resolution uses the sandbox's stdlib *socket.getaddrinfo*. Reverse DNS (PTR) uses *dig +short -x <ip> @1.1.1.1* against Cloudflare's public resolver. Both are cached so the pipeline is idempotent.

IP-to-ASN, country, and BGP-prefix attribution comes from Team Cymru's free bulk WHOIS service (*whois.cymru.com:43*), an industry-standard reference whose data is derived from public BGP routing tables. Every unique IP is queried in batches; results cached to *data/cache/asn_cymru.json*.

Cloud-region attribution for the three largest hyperscalers uses each vendor's own authoritative published IP-range manifest. GCP regions are read from *https://www.gstatic.com/ipranges/cloud.json* (941 prefixes, each tagged with a region scope such as *europe-west3*). Azure regions are read from Microsoft's *ServiceTags_Public_YYYYMMDD.json* published under *download.microsoft.com* (3,136 service-tag entries). AWS regions are extracted directly from the PTR record, which encodes the region by convention (*ec2-X-X-X-X.eu-west-1.compute.amazonaws.com*).

Sub-country location for OVH, Hetzner, Contabo, Leaseweb, and WorldStream comes from the MaxMind GeoLite2-City database, downloaded from a public mirror. GeoLite2 has known patchy coverage on some cloud blocks; cases where it returns only a country are flagged as such in the data.

Data from the diagnostic repository

The 901-pool / 16.40B ADA / 76.7%-productive-stake / 85-MPO-entity figures (the framing the original analysis question cited) come from the existing *mpo_entity_pool_mapping_mainnet.csv* and the diagnostic README, both already present in the repository before this analysis. The topology pass reads them rather than recomputing them.

The curated entity profiles — the “curated profile says” column in the cross-checks table — are verbatim quotes from *profiles/coinbase.md*, *profiles/everstake.md*, *profiles/kiln.md*, *profiles/blockdaemon.md*, and related files. These are documents written by the diagnostic team; the topology pass does not modify or interpret them, only sets the curated claim alongside what registered-relay data actually shows.

Curated reference tables encoded in the classifier

The provider classifier uses two static reference tables maintained in *build_mpo_topology_concentration.py*: an ASN-to-provider mapping (*AS16509* → *AWS*, *AS24940* → *Hetzner*, *AS16276* → *OVH*, *AS51167* → *Contabo*, *AS59642* → *Cherry Servers*, and others) and a PTR-pattern table (**.amazonaws.com* → *AWS*, **.your-server.de* → *Hetzner*, **.bc.googleusercontent.com*

→ *GCP*, **.cloudapp.net* → *Azure*, and others). These mappings are public conventions documented by each provider's own networking documentation and stable across years. They have not been fetched from a live authoritative source in this pipeline. For the major providers (AWS, GCP, Azure, OVH, Hetzner, Contabo, DigitalOcean) the mappings are well established. For the long-tail small hosters, the AS-name keyword fallback (*as_name* from Team Cymru) provides a cross-check.

Cloud-region geography (*eu-north-1 = Stockholm*, *ap-northeast-1 = Tokyo*, *europe-west3 = Frankfurt*, *eu-west-1 = Ireland*, and similar) is also encoded from public AWS / GCP / Azure documentation rather than fetched live.

Independent cross-validation

An independent measurement by Cexplorer.io (article: *Cardano infrastructure under scrutiny*) reports node-count shares of AWS 16%, Contabo 11%, DigitalOcean 10%, Google Cloud 7%, and self-hosted 6.4%. This pass reports stake-weighted shares of AWS 22.7%, OVH 11.9%, GCP 11.6%, Contabo 4.7%, DigitalOcean 2.4%. The differences between the two sources are **consistent and structurally explained**: AWS and GCP host institutional MPOs with concentrated stake, so their *stake* shares exceed their *node* shares; Contabo and DigitalOcean host many small single-pool operators with low individual stakes, so their *node* shares exceed their *stake* shares. This is exactly the asymmetric-cohort effect the MPO-vs-single-pool table makes explicit. Programmatic cross-validation against a public JSON-format topology tracker was not feasible from the Cowork sandbox (pooltool / adapools APIs are not currently allowlisted); this remains the highest-value follow-up.

Interactive map artefact

A companion interactive Leaflet map ships alongside this report at *outputs/mpo_topology_map.html*. It plots one pin per active pool with resolvable geography (2,429 of 2,915 pools), colour-coded by dominant provider and sized by log of active stake. Click any pin to see the entity, full provider mix, and registered relay regions. The provider filter on the side panel toggles individual provider categories on/off. The map opens directly in any modern browser.

Inferences drawn from the data

A small number of entity-level statements combine direct observation with publicly known background. The claim that Upbit sits almost exclusively in AWS Seoul derives from direct PTR observation (their relays resolve to *ap-northeast-2* endpoints) combined with public knowledge that Upbit is a South Korean exchange. The claim that Coinbase has the most resilient observed posture is computed: six distinct hyperscaler regions across two providers, with a corresponding Shannon-entropy diversity score of 0.802 nats in the profile-diff table. The OVH geographic spread (Roubaix, Strasbourg, Beauharnois, Frankfurt, Montréal) is from direct GeoLite2 city lookup.

Known limitations of the data

Three weaknesses warrant explicit acknowledgement before the numbers are used to inform a Leios-design decision.

The 26%-of-stake threshold used throughout the failure-mode analysis is taken directly from CIP-0164's own security analysis: *"a 26% stake attacker can trivially attack Leios throughput with a 75% certification threshold."* The CIP does not bind a numeric value for the certification threshold τ (it is left to governance); 75% is the illustrative value in the CIP. If governance selects a lower τ (e.g., 60%), the corresponding adversarial-stake threshold relaxes proportionally and the gap in the scenarios table widens. The committee-selection function (Persistent Voters via Fait Accompli sortition + Non-persistent Voters via Poisson-distributed local sortition) introduces additional sampling variance that is not modelled here; the stake-share threshold is therefore a necessary-but-not-sufficient condition for liveness, and the scenarios table reads as a hazard inventory rather than a quorum-failure probability computation.

The analysis describes registered relays, not block-producing nodes. Serious operators routinely hide producers behind public-facing relays in different facilities. The concentration figures are a **lower bound** on real operator-side concentration. Quantifying producer location from public data is non-trivial and out of scope for this pass; the most realistic route would be block-propagation timing analysis combined with a Cardano-node peer-state crawl.

Country aggregation uses the IP-registration country reported by Team Cymru, which for hyperscaler-hosted endpoints reflects the company's home country (Amazon / Google / Microsoft all US-registered) rather than the physical datacentre. Country-level figures are therefore noisy; region-level figures are the faithful read of physical location.

Reproducibility

Every artefact under *outputs/*, *figures/*, and *docs/* is regenerated by running, in sequence: *fetch_koios_pool_list.py*, *build_mpo_topology_concentration.py all*, *build_mpo_topology_sensitivity.py*, *build_mpo_topology_figure.py*, *build_mpo_topology_pdf.py*. The intermediate caches under *data/cache/* make the pipeline idempotent — re-running picks up only changed inputs. The full pipeline takes roughly five minutes against the public APIs used.

References

- *Specification:* ***CIP-0164 — Ouroboros Linear Leios*** *(cips.cardano.org/cip/CIP-0164)*
- *Scenario library: data/mpo_topology_failure_scenarios.csv (built by scripts/build_mpo_topology_scenarios.py)*
- *Tau-sensitivity: data/mpo_topology_tau_sensitivity.csv*
- *MPO vs single-pool split: data/mpo_topology_mpo_vs_spo.csv*
- *Concentration index (HHI / effective domains): data/mpo_topology_concentration_index.csv*
- *Interactive map (HTML, opens in any browser): outputs/mpo_topology_map.html (built by scripts/build_mpo_topology_map.py)*
- *Cexplorer independent comparison: cexplorer.io/article/cardano-infrastructure-under-scrutiny*
- *Extra-metrics script: scripts/build_mpo_topology_extra_metrics.py*
- *Physical-country table: data/mpo_topology_concentration_by_physical_country.csv*
- *Koios pool snapshot: data/koios_pool_list.json (built by scripts/fetch_koios_pool_list.py)*
- *MPO mapping: data/mpo_entity_pool_mapping_mainnet.csv*
- *Per-endpoint output: data/mpo_relay_endpoints_resolved.csv*
- *Concentration tables: data/mpo_topology_concentration_by_{provider,tier,country,region}.csv*
- *Entity × provider matrix: data/mpo_topology_entity_provider_matrix.csv*
- *Sensitivity analysis: data/mpo_topology_sensitivity.csv*
- *Entity profile diff: data/mpo_topology_entity_profile_diff.csv*
- *Findings narrative: docs/mpo_topology_findings.md*
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- *Sensitivity / profile-diff script: scripts/build_mpo_topology_sensitivity.py*
- *Figure script: scripts/build_mpo_topology_figure.py*
- *PDF script: scripts/build_mpo_topology_pdf.py*
- *ASN data: Team Cymru bulk WHOIS (whois.cymru.com:43)*
- *GCP IP ranges: gstatic.com/ipranges/cloud.json*
- *Azure ServiceTags: download.microsoft.com (Microsoft public JSON)*
- *GeoLite2-City: MaxMind (free database via public mirror)*